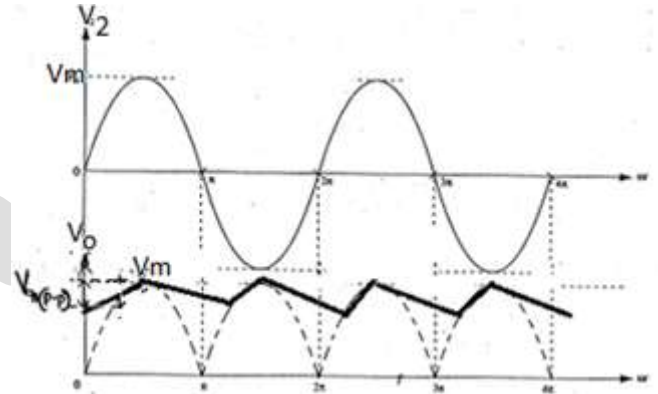
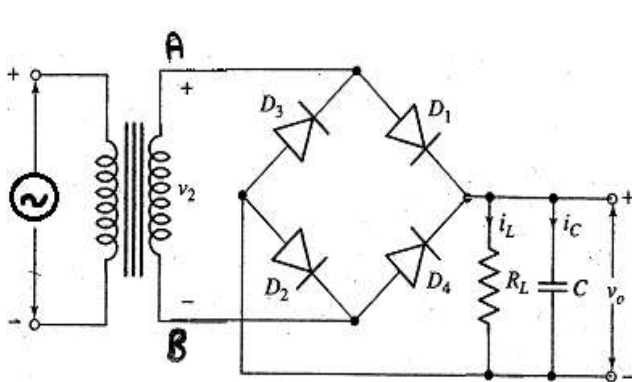


Q1. Explain the operation of Full Wave Bridge Rectifier with C filter

Operation: During the +ve half cycle of the input, the diodes D_1 and D_3 are forward biased and D_2 and D_4 are reverse biased, therefore D_1 and D_3 starts conducting. Since the capacitor acts as a short circuit for ac signal, it gets charged up to the peak of the input signal V_m and the dc component flows through the load R_L .



When the input signal falls below V_m , the diodes D_1 and D_3 gets reverse biased. Now the charged capacitor acts as a battery and it starts discharging through the load R_L till the diodes D_2 and D_4 starts conducting in the -ve half cycle. When diodes D_2 and D_4 starts conducting, the capacitor starts charging to the voltage V_m .

When the input signal falls below V_m , the diodes D_2 and D_4 are now reverse biased. Now the charged capacitor again acts as a battery and now it starts discharging again through the load R_L till the Diode D_1 and D_3 starts conducting in the +ve half cycle and the process repeats. From the waveforms, it can be seen that without filter, the output voltage varies between zero and V_m . With Capacitor filter, the output voltage varies between $[V_m - V_{r(p-p)}]$ and V_m . Hence shunting of load resistor R_L by capacitor C reduces the ripple content in the output.

The ripple factor with C filter for FWR is given by $\gamma = \frac{1}{4\sqrt{3} f C R_L}$

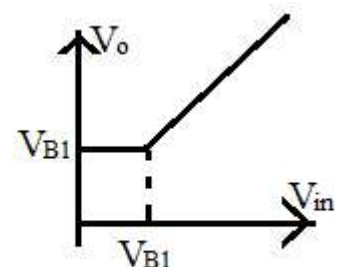
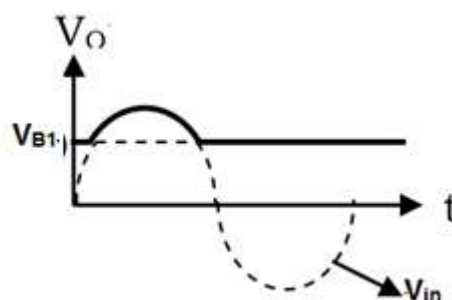
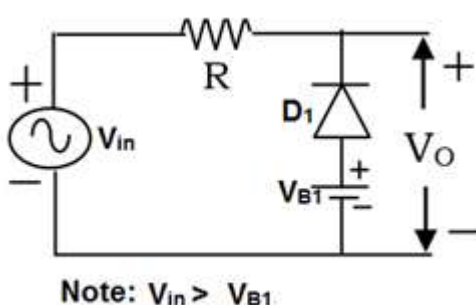
What are Clippers? Explain the operation of (i) single level clipper (ii) two level clipper

The function of the clipper is to remove the unwanted portion of the wave form. Clippers are used as protective devices in circuits against large amplitude variations. A half wave rectifier acts a clipping circuit which remove the negative half of the ac input signal.

Single level Shunt Clippers: The diode is connected across the output .

Shunt clippers - Clipping below the reference level

Figure shows the circuit diagram input/output waveform and transfer characteristics of shunt clipper used clip below the reference level V_{B1} . The diode is assumed to ideal.



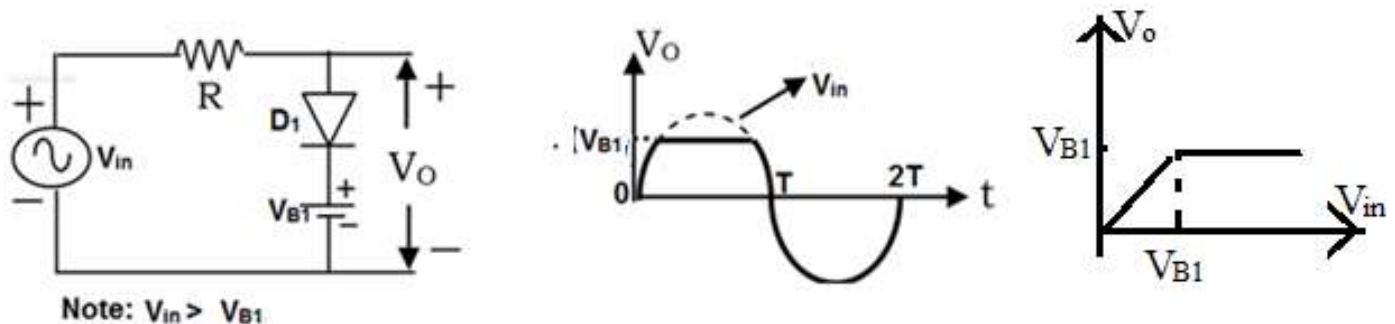
Consider the positive half cycle of the input, the diode D_1 is initially forward biased and the output voltage is at V_{B1} till the input voltage exceeds this level. $V_O = V_{B1}$

Once the input voltage is greater than V_{B1} , the diode is reverse biased and output follows the input and the hence the output voltage $V_O = V_{in}$

In the negative half cycle of the input, the forward biased diode is further forward biased and the output is held at $V_O = V_{B1}$

Shunt clippers - Clipping above the reference level

Figure shows the circuit diagram input/output waveform and transfer characteristics of shunt clipper used clip above the reference level V_{B1} . The diode is assumed to ideal.



Consider the positive half cycle of the input, the diode D_1 is initially reverse biased and acts as an open circuit. Hence the input is passed on to the output till the input voltage is less than V_{B1} .

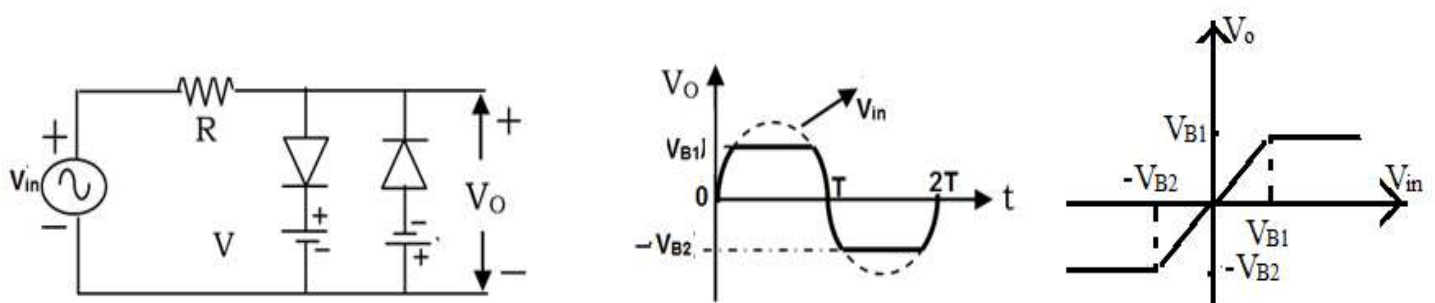
Once the input voltage crosses this level. The diode is forward biased and output voltage will be $V_O = V_{B1}$

In the negative half cycle of the input, the reverse biased diode is further reverse biased and the input is passed on to the output and $V_O = V_{in}$

(ii) Two level Clippers (Combinational clipper)

Figure shows the circuit diagram of a biased shunt clipper. The reference voltages V_{B1} and V_{B2} are used for biasing the diodes. The diodes are assumed to be ideal. The diodes are initially reverse biased.

Analysis: During the positive half cycle of the input, till the applied voltage is less than V_{B1} , both the diodes are reverse biased and output voltage is equal to input voltage.



When the applied voltage is greater than V_{B1} , the diode D_1 is forward biased and diode D_2 which was already reverse biased by the supply V_{B2} gets further reverse biased.

Now the output voltage is $V_O = V_{B1}$

In the negative half cycle of the input, till the applied voltage is less than $-V_{B2}$, both the diodes are reverse biased and output voltage is equal to input voltage.

When the applied voltage is greater than $-(V_{B2})$, the diode D_2 is forward biased and diode D_1 which was already reverse biased by the supply V_{B1} gets further reverse biased.

Now the output voltage is $V_O = -(V_{B1})$.

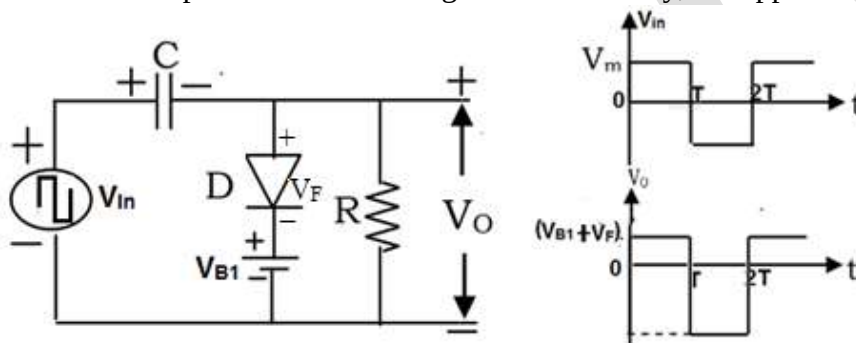
Explain (i) Positive Clamping Circuit and (ii) Negative clamping circuit with reference:

Clamper circuits are called DC storer circuits and are used to shift the dc level of the waveform without affecting the shape of the waveform (peak to peak voltage of the waveform remains constant).

Biased Negative Clamper- Figure shows a biased positive clamper with reference voltage V_{B1} . The value of the resistor selected is very high so that the time constant $RC > T$ time period of the input signal. Low values of R , will produce a tilt in the output waveform.

Analysis of the circuit

The diode is assumed to be ideal and initially reverse biased. During positive half cycle of the input ($0 < t < T$) as the voltage across the capacitor can not change instantaneously, the applied voltage forward biases the diode.



Hence the output voltage is $V_O = V_{B1}$

Now the capacitor charges to a potential V_C with the polarity as shown and is given by

$$V_{in} - V_C - V_{B1} = 0 \quad \text{or} \quad V_C = [V_m - V_{B1}] \quad \text{where } V_{in} = V_m$$

During the negative half cycle of the input ($T < t < 2T$), the diode gets reverse biased, now the capacitor discharges thro' the resistor R . The output voltage is given by

$$V_{in} - V_C - V_O = 0 \quad \text{or} \quad V_O = V_{in} - V_C \quad \text{substituting the values for } V_{in} = -V_m \text{ and } V_C, \text{ we get}$$

$$V_O = -V_m - (V_m - V_{B1}) = [-2V_m + V_{B1}]$$

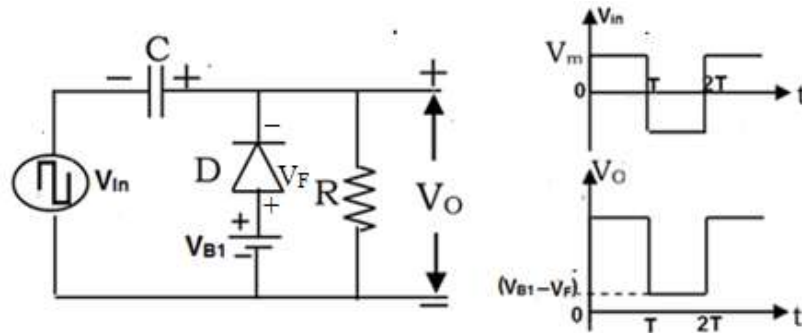
The peak to peak output voltage is given by $(V_{B1}) - (-2V_m + V_{B1}) = 2V_m$

Biased Positive Clamper:

Figure shows a biased positive clamper with reference voltage V_{B1} . The value of the resistor selected is very high so that the time constant $RC > T$ time period of the input signal. Low values of R , will produce a tilt in the output waveform.

Analysis of the circuit - The diode is assumed to be ideal and initially forward biased. During negative half cycle of the input period ($T < t < 2T$), as the voltage across the capacitor can not change instantaneously, the applied voltage further forward biases the diode.

Hence the output voltage is $V_O = (V_{B1})$



Now the capacitor charges to a potential V_C with the polarity as shown and is given by

$$V_{in} + V_C - V_O = 0 \quad \text{or} \quad V_C = -[V_{in} - V_O] \quad \text{substituting } V_{in} = -V_m \text{ and } V_O = (V_{B1})$$

$$V_C = -[-V_m - V_{B1}] = [V_m + V_{B1}]$$

During the positive half cycle of the input ($0 < t < T$), the diode gets reverse biased, now the capacitor discharges thro' the resistor R. The output voltage is given by

$$V_{in} + V_C - V_O = 0 \quad \text{or} \quad V_O = V_{in} + V_C \quad \text{substituting the values for } V_{in} = V_m \text{ and } V_C, \text{ we get}$$

$$V_O = V_m + [V_m + V_{B1}] = [2V_m + V_{B1}]$$

The peak to peak output voltage is given by $(2V_m + V_{B1}) - (V_{B1}) = 2V_m$

Q. Explain the operation of precision rectifier

Precision Rectifiers

The normal rectifiers using diode cannot rectify voltages below 0.7V (the cut in voltage of the diode).

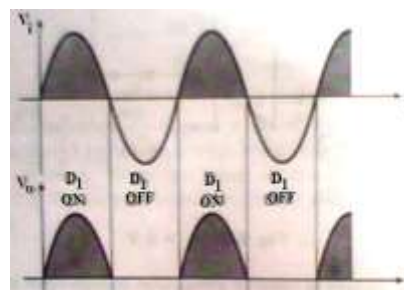
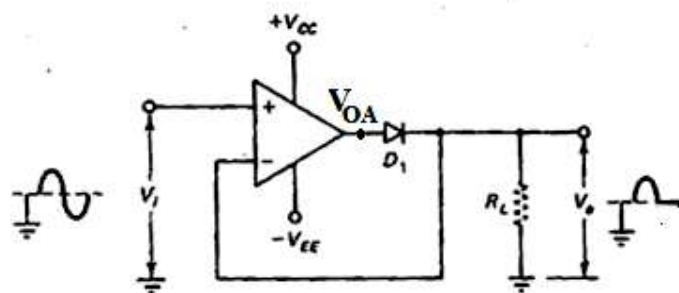
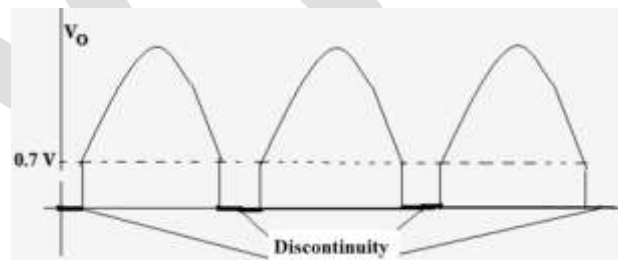
Hence $V_O = V_i - V_{D(ON)} = V_i - 0.7$ for $V_i > V_{D(ON)}$

$V_O = 0$ for $V_i < V_{D(ON)}$. Hence the output is discontinuous and distorted.

An op-amp can be used along with diodes to achieve precision rectification wherein V_O is continuous for all values of $V_i > 0$ and these circuits are called precision rectifiers.

Precision Half Wave Rectifiers

Fig shows a +ve half wave precision rectifier, the circuit is basically a voltage follower with a diode D_1 inserted in the feedback path of the circuit.



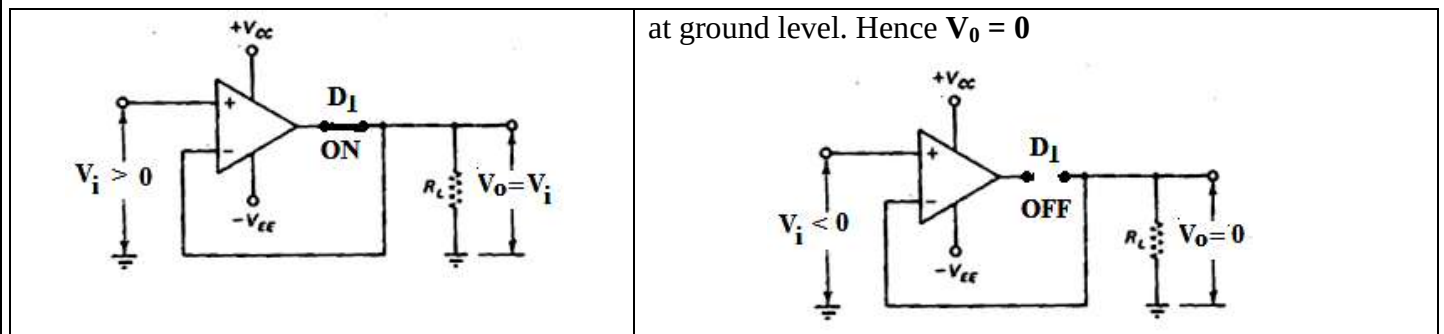
Analysis

When the input signal is positive $V_i > 0$;

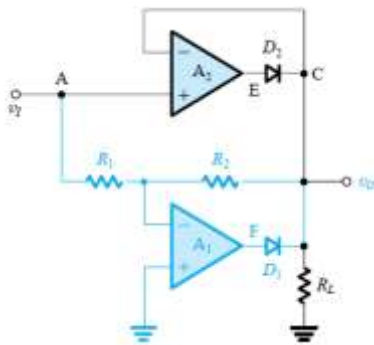
due high open loop gain of the op-amp it produces a high voltage V_{OA} which in-turn makes the diode forward biased and the output voltage follows the input. Hence $V_O = V_i$

When the input signal is negative $V_i < 0$;

the voltage V_{OA} attains $-V_{Sat}$ value making the diode D_1 reverse biased, hence the diode acts as an open circuit. So the feedback path gets open, since R_L is grounded which in-turn makes the inverting terminal



Precision Full Wave Rectifier – Figure shows the circuit diagram of precision full wave rectifier. The circuit consists of a precision HWR and an inverting amplifier having a diode in the forward path, and the two outputs are merged to generate common output.



Operation – During the positive half cycle of the input $v_i > 0$; diode D_1 is forward biased and D_2 is reverse biased. Op-amp A_1 output is zero and opamp A_2 acts as voltage follower. Hence the output will be same as input $v_o = v_i$.

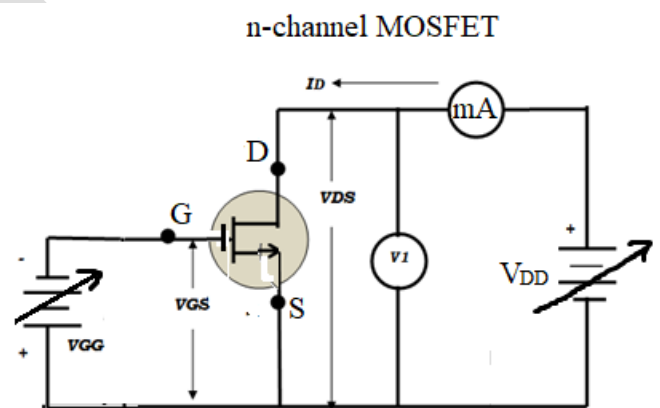
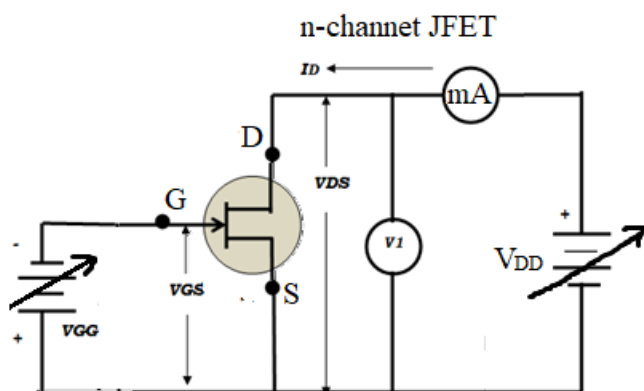
During the negative half cycle of the input $v_i < 0$; diode D_1 is forward biased and D_2 is reverse biased. Output of A_2 is zero and the opamp A_1 acts as an inverting amplifier. Hence the output will be $v_o = -\left(\frac{R_2}{R_1}\right)(-v_i) = \left(\frac{R_2}{R_1}\right)(v_i)$

For $R_2 = R_1$, then $v_o = v_i$.

Hence the total output is a precision full rectified sine wave for one full cycle of the input.

Obtain the Transfer and Drain (output) Characteristics of JFET/ MOSFET

Figure shows the circuit diagram of n-channel JFET/ MOSFET used for determining its o/p and transfer characteristics.

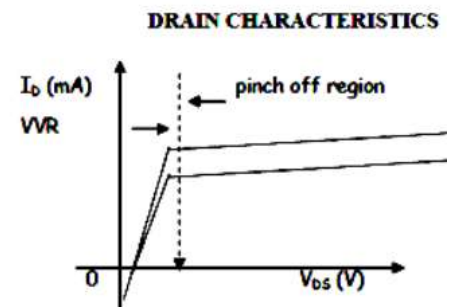


Procedure:

Drain Characteristics:

1. Connect the circuit as shown in the figure
2. Keep $V_{GS} = 0V$ by varying V_{GG} .
3. Varying V_{DD} gradually in steps of 1V up to 10V note down drain current I_D and drain to source voltage (V_{DS}).
4. Repeat above procedure for $V_{GS} = -0.4, -0.8, -1.2$ and $-1.6 V$
4. Plot the graph of I_D vs V_{DS} or different values of V_{GS}

Drain Characteristics						
V_{DS} (Volts)	$V_{GS} = 0V$		$V_{GS} = -0.4V$		$V_{GS} = -0.8V$	$V_{GS} = -1.2V$
	$V_{DS}(\text{Volts})$	$I_D(\text{mA})$	$V_{DS}(\text{Volts})$	$I_D(\text{mA})$	$V_{DS}(\text{Volts})$	$I_D(\text{mA})$
0						
2						
4						
6						
8						
10						

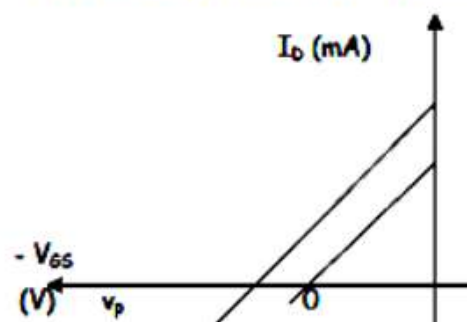


Transfer Characteristics:

1. Connect the circuit as shown in the figure.
2. Set voltage $V_{DS} = 4V/8V$
3. Varying V_{GS} in steps of $0.5V$ until the current I_D reduces to minimum value.
4. Varying V_{GG} gradually, note down both drain current I_D and gate-source voltage (V_{GS}).
5. Repeat above procedure (step 3) for $V_{DS} = 4V/ 8V$
6. Plot the graph of I_D vs V_{GS} at constant V_{DS}

Transfer Characteristics			
$V_{DS} = 4V$		$V_{DS} = 8V$	
V_{GS} (Volts)	I_D (mA)	V_{GS} (Volts)	I_D (mA)
0			
0.5			
1.0			
1.5			

TRANSFER CHARACTERISTICS



Calculations from Graph:

1. Drain Resistance (r_d):

$$r_d = \frac{\Delta V_{DS}}{\Delta I_D} \text{ at a constant } V_{GS} \text{ (from drain characteristics)}$$

2. Trans Conductance (g_m):

$$g_m = \frac{\Delta I_D}{\Delta V_{GS}} \text{ at constant } V_{DS} \text{ (from transfer characteristics).}$$

The value of g_m is expressed in mho's (Ω) or Siemens (s).

3. Amplification factor (μ):

It is given by the ratio of small change in drain to source voltage (ΔV_{DS}) to the corresponding change in gate to source voltage (ΔV_{GS}) for a constant drain current (I_D).

$$\mu = \left(\frac{\Delta V_{DS}}{\Delta I_D} \right) \times \left(\frac{\Delta I_D}{\Delta V_{GS}} \right) = \frac{\Delta V_{DS}}{\Delta V_{GS}}$$

$$\text{ie. } \mu = r_d \times g_m$$

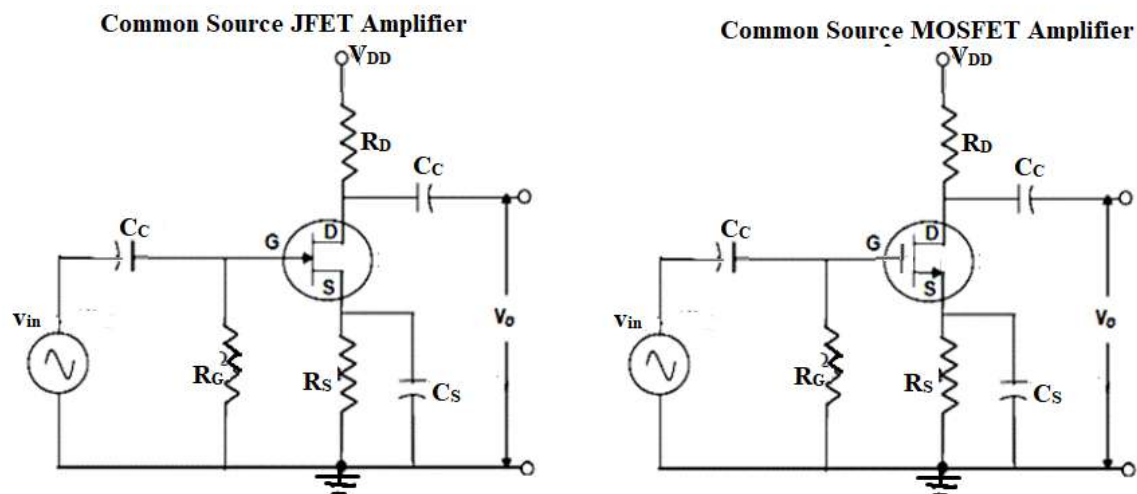
Explain the frequency response of Common source amplifier using JFET/MOSFET.

Frequency response of an amplifier is graph of output voltage for different values of input signal frequencies keeping the input signal amplitude constant.

Procedure: 1. Connect the circuit as shown in the circuit diagram.

2. Set source voltage $V_{in} = 50mV$ (say) at 1 KHz frequency using the function generator.

3. Keeping input voltage constant vary the frequency from 50 Hz to 1 MHz in regular steps and note down the corresponding output voltage.



4. Plot the graph: gain (dB) versus Frequency on a semi log graph sheet.

5. Calculate the bandwidth from the graph. 6. Calculate all the parameters at mid band frequencies

$$\text{Voltage gain } A_v = \frac{\text{Output voltage } v_o}{\text{Input voltage } v_{in}} \quad \text{or} \quad \text{Gain in dB} = 20 \log \left(\frac{v_o}{v_{in}} \right) \quad \text{and} \quad BW = [f_H - f_L] \quad \text{in Hz}$$

Cut-off frequencies are the frequency at which gain drops by 3 dB from its maximum value and the difference between cut-off frequencies gives the bandwidth of the amplifier

Tabular Column $V_i = 50\text{mV}$

Frequency	Vo(volts)	Gain= Vo/Vs	Gain(dB)=20 log(Vo/Vs)

Frequency Response